TATA CONSULTANCY SERVICES

SPEC Driven Development for SDLC using AI

TATA

AI Fridays

DURATION: 5 HOURS

DEVELOP SOFTWARE USING ARTIFICIAL INTELLIGENCE (AI) TOOLS/AI Agents FOR THE FOLLOWING USE STORY

Evaluation Criteria (Read this, Very Important)

Evaluation Criteria

The evaluation process will focus on the quality of each deliverable outlined in this document. It is important to note that assessment goes beyond merely ensuring the software functions as intended. The effectiveness of AI tools and agents, as integrated across every phase of the Software Development Life Cycle, will be a primary consideration.

Every deliverable should be comprehensive, presented in a professional manner, and show clear completion. It is recommended to use AI tools to review all deliverables before finalizing them.

Maintain a comprehensive record of all prompts utilized for the creation of various deliverables. Ensure that you are logged into your tool during use so prompt history remains readily accessible.

Key Areas of Assessment

- **Comprehensive Application of AI**: The use of AI in analysis, design, development, testing, and documentation must be demonstrated for all required outcomes.
- **Impactful Implementation:** Evaluators will look for clear evidence of how AI technologies have positively influenced each stage of the software development process.
- **Deliverable Quality and Completeness**: Each deliverable must be thoroughly developed, meeting all specified requirements with high quality and completeness.

- **Innovative Approach:** Out-of-the-box thinking will be valued, especially in the creation of deliverables. For example, submitting a Functional Specification Document in HTML format instead of the traditional Word document format will be considered innovative.

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INDICATIVE LIST OF TOOLS THAT CAN BE USED

AI App & Workflow Builders

- Lovable AI (Free tier)
- Bolt.new (Free credits)
- Replit (Free tier with AI features)
- Vercel v0 (Free tier)

LLM & Agent Development Platforms

- Google AI Studio (Gemini)
- OpenAI Playground (Free credits)
- Azure AI Studio (Free tier)
- Hugging Face Spaces
- LangChain (Open source)
- LlamaIndex (Open source)
- CrewAI (Open source)
- AutoGen (Open source)

AI Coding & Code-Quality Agents

- Codeium (Free)
- Cursor (Free tier)
- Continue.dev (Open source)
- Semgrep (Free tier)
- SonarQube Community Edition

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AI for Requirements, Documentation & Analysis

- ChatGPT (Free)
- Claude (Free tier)
- Notion AI (Limited free usage)
- Obsidian + community AI plugins

AI for UI / UX & Wireframe Design

- Lovable AI (Free tier) ✅ *(explicitly included)*
- Uizard (Free tier)
- Figma (Free tier with AI features)
- Penpot (Open source)
- Diagrams.net (Draw.io AI)
- Whimsical (Free tier)

AI for Testing & QA

- Postman AI (Free tier)
- Playwright + AI plugins
- Selenium IDE + AI extensions
- OpenTest (Open source)

AI for Architecture, Data Modeling & Diagrams

- DBdiagram.io (Free tier)
- Creately AI (Free tier)
- PlantUML (AI-assisted via prompts)

AI Agent UI / Demo Frameworks

- Streamlit (Open source)
- Gradio (Open source)

- Chainlit (Open source)

AI Search & Research

- Perplexity AI (Free)
- Phind (Free)
- You.com (Free AI modes)

Local / Open-Source Models & Runtimes

- Ollama
- LM Studio
- Mistral (Open models)
- Groq (Free tier)

SPEC generation

- SpecGPT (spec generation with AI)
- OpenAI GPT-4 (prompted for requirement/spec generation)
- Notion AI (automated documentation and spec drafting)
- Jira (with AI plugins for converting user stories to specs)
- Confluence (integrated AI tools for technical documentation)
- GitHub Copilot (for generating code-ready specifications)
- Mintlify (AI-powered spec and documentation generator)
- DocuWriter.ai (automated technical spec writing)

User Case

AI-Assisted Service Deviation & Approval Workflow (Telecom & Media)

Operational Deviation Initiation and Approval Workflow

User Roles and Responsibilities

Telecom or Media Operations Managers, Network Managers, and Account Managers are empowered to initiate and manage operational deviations. These deviations may include actions such as granting billing credits that exceed standard limits, approving temporary increases in data or bandwidth, permitting limited SLA waivers, allowing content access exceptions, or authorising short-term KYC deferrals. Each request is processed through a controlled, multi-level approval workflow that involves various teams, including Operations, Finance, Network, and Compliance. The objective is to ensure that customer requirements are met efficiently while maintaining strict adherence to regulatory guidelines and revenue control mechanisms.

Workflow Controls and Compliance Measures

The operational deviation workflow is designed to mandate a clear business justification for each deviation, along with all necessary checks and documented approvals. AI capabilities are integrated to assist by generating standardised, compliance-friendly justifications, referencing relevant internal policies or contractual clauses, and summarising customer and service context for decision-makers. The process enforces role-based approvals, ensuring that mandatory compliance steps cannot be bypassed and that execution occurs only after receiving final approval. All actions and decisions taken within the workflow are systematically logged, producing audit-ready records that support both internal audits and external regulatory reviews.

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Deliverables

All the deliverables are expected to be of production-level quality, ensuring excellence in both content and look & feel.

1. **A functional requirements document (or user story set)** - AI-assisted service deviation and approval workflow should clearly define the end-to-end business flow and system Behavior, covering key elements such as business objectives and scope, actors and roles (requestors, approvers, reviewers, AI agents), deviation types and triggers, detailed workflow states and transitions, approval hierarchies and escalation rules, AI-assisted capabilities (recommendations, risk scoring, policy interpretation, explanation/justification), decision override and human-in-the-loop controls, input/output data elements, validation rules, auditability and traceability requirements, exception handling and fallbacks, integrations with upstream/downstream systems, notifications and SLAs, security and access controls, compliance constraints, and reporting/analytics needs. Each requirement should be unambiguous, testable, and mapped to business outcomes to ensure both automation and governance are addressed comprehensively.
2. **SPECS & DSL** – Create Markdown language or YAML for each user story component, which will be converted to code.
3. Application Architecture and technical architecture diagrams
4. Database design / Data Models – No JSON input of data, use any RDBMS, deliverables include logical model, physical model and ER diagram
5. Software code & Working Software
6. Functional Test scripts in a CSV or Excel
7. Traceability Matrix in CSV or Excel
8. End User manual - Screens and help guide

Instructions

Use AI tools or Build Your Own AI Agents (BYOAA) to develop the software and deliverables.

- **Selection of Tools:** Carefully evaluate and select the most suitable AI tools or frameworks for each phase of the project. If building custom AI agents, ensure they are scalable, maintainable. Additional weightage will be given if team builds their own AI agent to create the deliverables.

- **Quality Standards:** Adhere strictly to coding standards and best practices for software development. Apply consistent design patterns, code reviews, and automated testing to maintain high quality.
- **Integration:** Ensure seamless integration of components with the overall application architecture.
- **Testing:** Prepare comprehensive test cases, including both functional and non-functional scenarios.
- **Documentation:** Maintain clear and detailed documentation for all deliverables, including architecture diagrams, data models, and user manuals. Update documents regularly as changes are made.
- **User Experience:** Prioritise a user-centric approach in UI/UX design. Avoid using Streamlit or Gradio for the user interface.